

Jiaqi (Victoria) Duan

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EDUCATION

University of California, Santa Cruz

Computer Science and Engineering, Master of Science (M.S.) Dec 2025
Computer Science and Engineering, Bachelor of Science (B.S.); Psychology, Bachelor of Art (B.A.) Dec 2022

SKILLS

- **Programming Languages:** TypeScript, JavaScript (ES6+), Python, Java, C/C++
- **Frontend Development:** React, React Native (Expo), Next.js, Redux Toolkit, Tailwind CSS, NativeWind, Vite, HTML5, CSS3, Sass, SolidJS, Axios
- **Backend & Databases:** FastAPI, Django, Express.js, Node.js, GraphQL, RESTful API & gRPC design, PostgreSQL, Firebase (Firestore), Supabase, Redis, Kafka, RabbitMQ, Microservices Architecture
- **Infrastructure & Cloud:** Google Cloud Platform (GCP), AWS, Docker, Kubernetes, Terraform, Vercel, Postman
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly, Chart.js, Logging & Monitoring
- **Generative AI & Tooling:** PyTorch, TensorFlow, SpaCy, OpenAI API, Gemini API, Claude, Hugging Face Transformers, Ollama, AutoGen, LangChain, LlamaIndex, Chroma, Pinecone, LastMile AI, Weights & Biases (W&B), Cursor

EXPERIENCE

Founding Engineer @ Ripplet | June 2024 – Aug 2025

- Architected and implemented a **HIPAA-compliant, multi-agent system** with **retrieval-augmented generation (RAG)** pipelines to analyze session transcripts and map client narratives to evidence-based psychological frameworks
- Built responsive dashboards with **Chart.js** to visualize **real-time** client sentiment trends, progress milestones, and engagement levels, helping therapists review sessions more effectively and tailor future treatment plans
- Developed and deployed Ripplet’s marketing site with **Next.js**, **React**, **Tailwind CSS**, and **Vercel**, achieving pixel-perfect brand consistency, **responsive design**, and optimized performance with **SEO** best practices

Full Stack Engineer @ Tech4Good Lab | June 2023 – Jan 2025

- Led a **cross-functional** team of engineers and designers to develop Pathways, an **AI-powered** self-directed learning platform, coordinating backend integration and pilot testing with **over 100 university students**
- Delivered the core application using **Solid.js**, **Express.js**, and **Firebase**, implementing **server-side rendering**, **lazy loading**, and **cached API responses**, **reducing average page load times by 35%** in staging tests
- Designed and refined LLM prompts with **different prompting strategies**, validated via **A/B testing**, **reducing hallucinations** and **improving course recommendation relevance by 15%** as measured by completion rates

PROJECTS

Travel Agent | Python, AutoGen, Redis

- Designed and built a **centrally orchestrated multi-agent system** integrating **open-domain** retrieval, structured API search, itinerary synthesis, critic-driven refinement, and system-level fallback in a deterministic pipeline
- Built a session-scoped shared state layer using **Redis** to manage cross-agent context, caching retrieved evidence, intermediate artifacts, and critique feedback to support consistent **multi-step reasoning**
- Implemented a **critic-driven self-improvement loop** where generated itineraries are evaluated for factuality, feasibility, constraint satisfaction, safety, and currency, and iteratively revised until acceptance or a retry limit
- Conducted a four-way ablation study on real user cases, demonstrating that critic-guided feedback improves itinerary quality, with **critic scores moderately correlated with human judgments** (Pearson $r \approx 0.54$)

Yi | React, Next.js, Django, RESTful API, PostgreSQL

- Built a **B2B SaaS application** for small businesses to unify business management and financial analytics, featuring interactive dashboards and visualizations for real-time performance tracking
- Engineered a **reusable component library** with built-in **theme switching** using **Tailwind CSS** configuration, ensuring consistent UI patterns, improving codebase **maintainability**, and facilitating faster feature development
- Developed **authentication** and **role-based access control (RBAC)** with **Django REST Framework** and **JWT**, enabling differentiated permissions for staff, admins, and guests to securely access relevant data and features
- Integrated **Redux Toolkit** and **RTK Query** for centralized **state management** and efficient backend integration, utilizing **caching** and **request deduplication** to improve performance and support scalable data flows
- Implemented a secure API layer for session data ingestion and retrieval, optimizing database access with **PostgreSQL** indexing and **Redis** caching, **reducing query latency by 120ms (30%)** in testing compared to unoptimized baselines